

Student Name:

Reg/ID #:

Multiply 101 with 101010 assuming that the product holds a value of 000000000 at the initialization/start. Use the sequential multiplication algorithm and show all the steps. Write legibly!

Sol:

Itr/cc	Step	Multiplier	Multiplicand	Product
0	Init	101	0 0010 1010	0 0000 0000
1	Prod + Mcnd Shift left Shift right	101 101 010	0 0010 1010 0 0101 0100 0 0101 0100	0 0010 1010 0 0010 1010 0 0010 1010
2	No opr Shift left Shift right	010 010 001	0 0101 0100 0 1010 1000 0 1010 1000	0 0010 1010 0 0010 1010 0 0010 1010
3	Prod + Mcnd Shift left Shift right	001 001 000	0 1010 1000 1 0101 0000 1 0101 0000	0 1101 0010 0 1101 0010 0 1101 0010